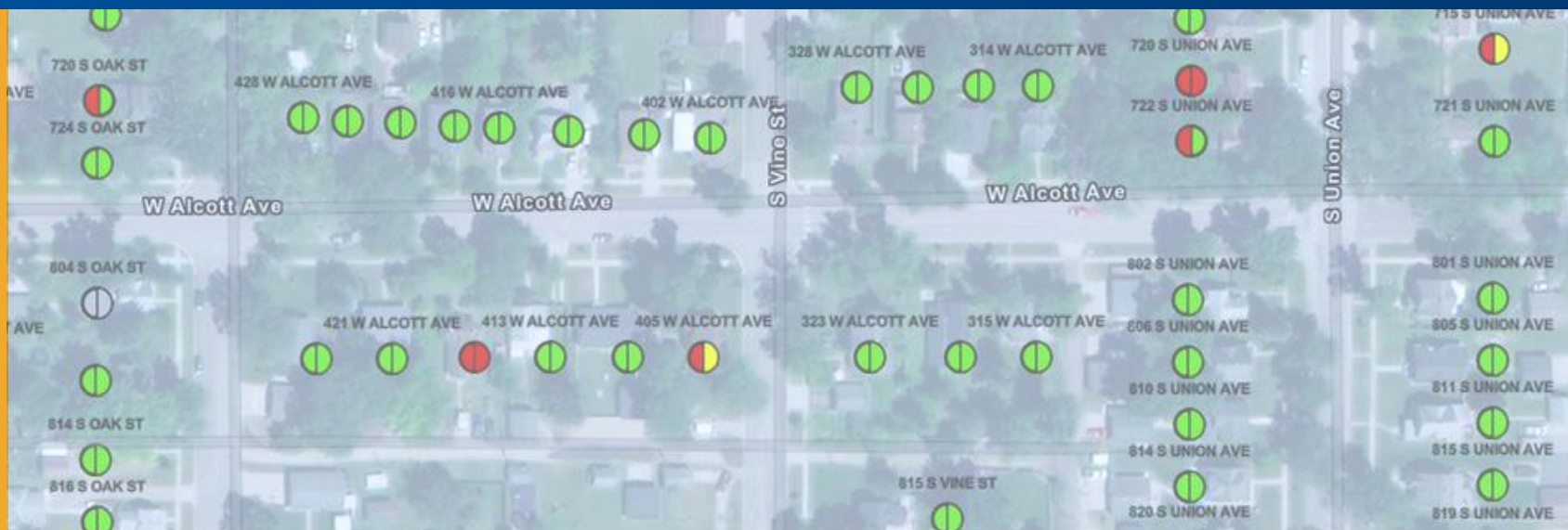


April 9, 2026

Amanda Hoffman
GIS Analyst II

2026 – Lunch and Learn

Lead Service Line Replacement GIS Applications



Lead Service Line (LSL): Background

- Lead and Copper Rule revisions require communities to identify service line materials across their system
- Replacement expectations are increasing, with long-term compliance deadlines now more clearly defined
- For small communities, it be challenging to track, coordinate, and communicate with all necessary stake-holders
- **Apex used GIS to help organize that work from inventory through replacement preparation**

Lead Service Line (LSL) Inventory

- Apex assisted 15 cities in 2025 with their LSL inventories, primarily small cities with no internal GIS staff.
- Work included reviews of engineering and utility records, mailed notices to land-owners, door to door surveys, and utility locating.
- **Problem: Needed a simple tool for project managers to review progress and QC inventory, GIS staff to update data, and city staff to contribute knowledge.**

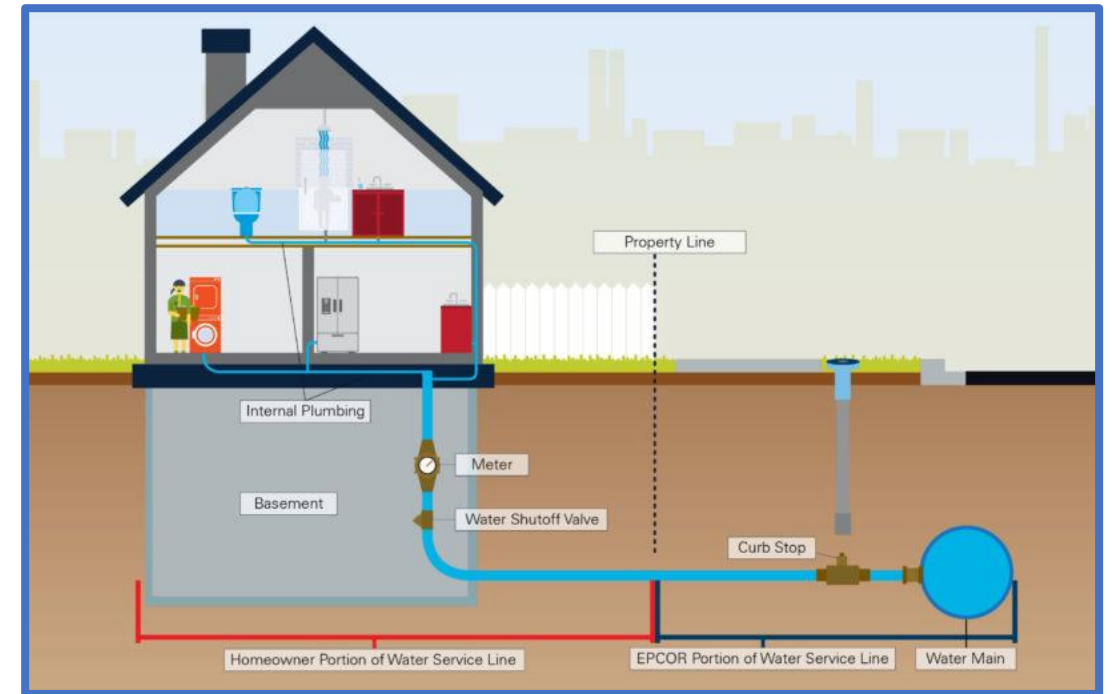
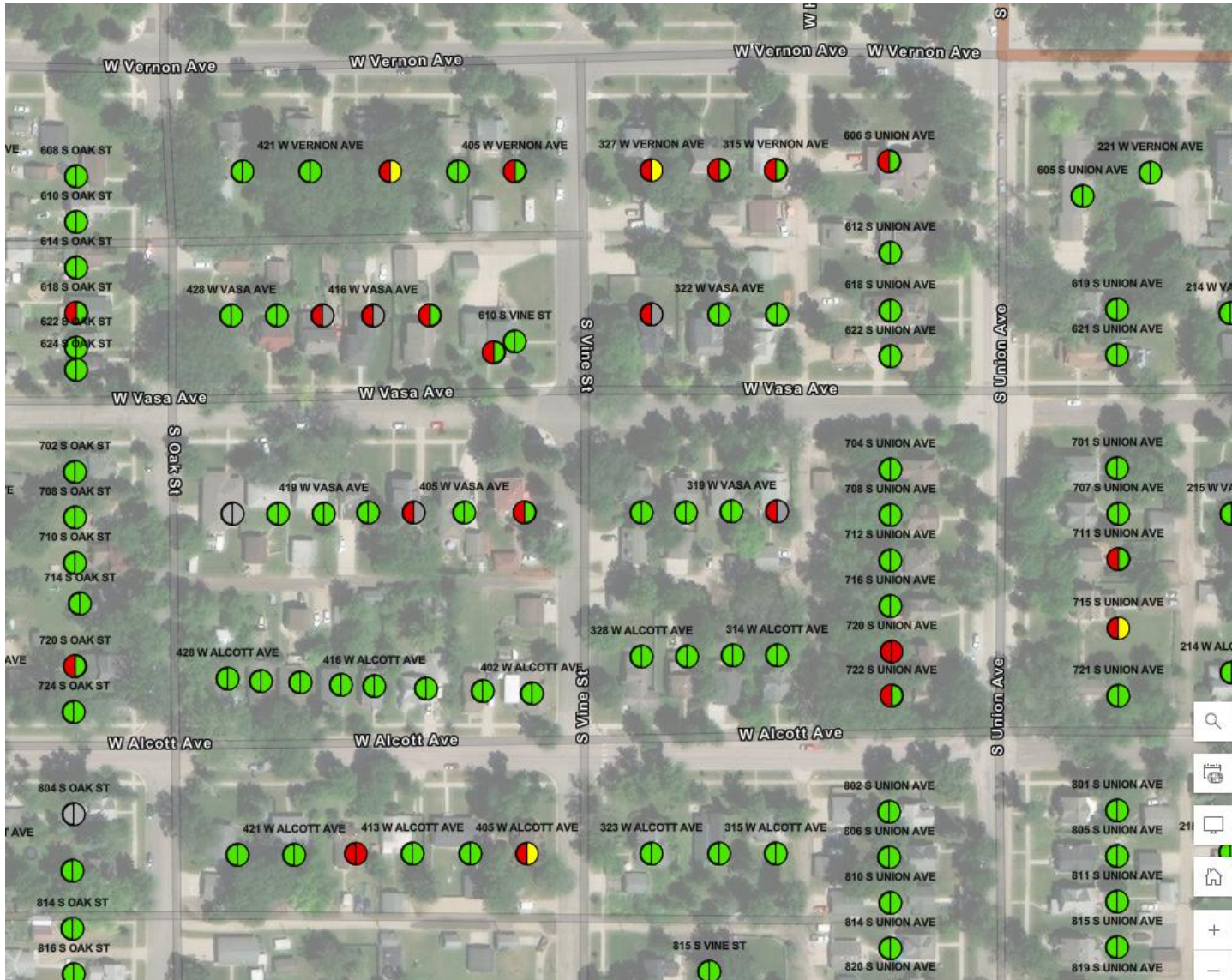


Image courtesy of EPCOR

LSL Inventory: Mapping



- GIS provided a parcel-level view of service line material status: non-lead, galvanized, lead, and unknown
- Mapping helped prioritize follow-up, identify data gaps, and communicate progress with city staff

Green – Non-Lead
Yellow – Galvanized
Red – Lead
Gray – Unknown

LSL Inventory: Material Identification

- Mailed Notices and Online Survey
- Door Knocking
- Utility Record Review

City of Pelican Rapids - Water Service Line Material Survey

Fields with red asterisk are required fields, remaining are optional.

Name*

Phone Number

() -

Email Contact

Property Address for Water Service*

Additional Location Information

Party Completing This Survey (Please Select One)*

☐ Owner

☐ Renter / Tenant

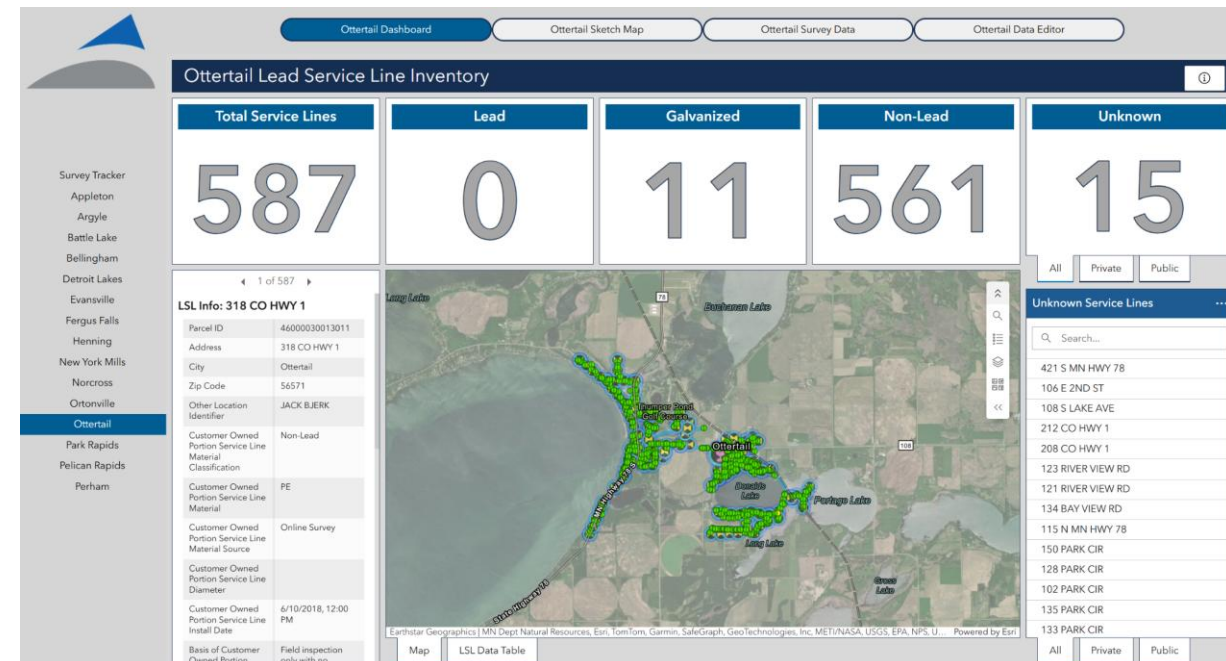
☐ Property Manager

LSL Inventory: Dashboards

■ Dashboards for Project Managers and GIS Staff

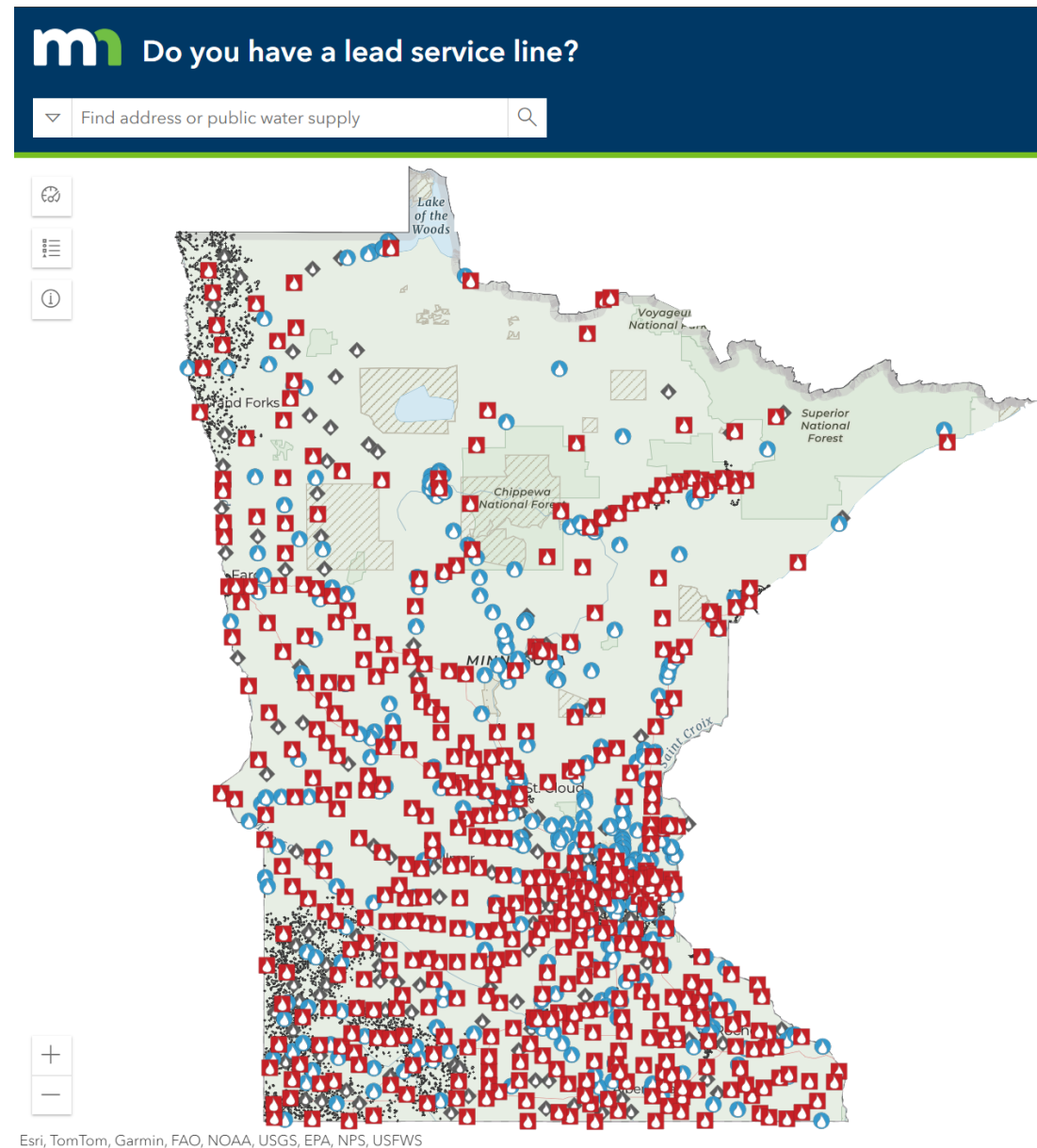
- Each city had a dashboard for tracking status, unknowns, and follow-up needs
- PMs could monitor progress and review QC issues
- GIS staff and city staff could update information in one shared system

Survey Tracker Appleton Argyle Battle Lake Bellingham Detroit Lakes Evensville Fergus Falls Henning New York Mills Norcross Ortonville Ottertail Park Rapids Pelican Rapids Perham	LSL Submitted Status Surveys					
	Argyle Online Survey	Argyle Door to Door Survey	Appleton Online Survey	Appleton Door to Door Sur...	Battle Lake Online Survey	Battle Lake Door to Door Sur...
	0	0	0	0	0	0
	Bellingham Online Survey	Bellingham Door to Door Sur...	Detroit Lakes Online Survey	Detroit Lakes Door to Door ...	Evensville Online Survey	Evensville Door to Door Surv...
	0	0	0	No Data	0	0
	Fergus Falls Online Survey	Fergus Falls Door to Door S...	Henning Online Survey	Henning Door to Door Survey	New York Mills Online Survey	New York Mills Door to Door...
	No Data	2	0	0	0	0
	Norcross Online Survey	Norcross Door to Door Survey	Ortonville Online Survey	Ortonville Door to Door Sur...	Ottertail Online Survey	Ottertail Door to Door Survey
	0	0	0	0	0	0
	Park Rapids Online Survey	Park Rapids Door to Door S...	Pelican Rapids Online Survey	Pelican Rapids Door to Doo...	Perham Online Survey	Perham Door to Door Survey
	0	0	1	0	0	0



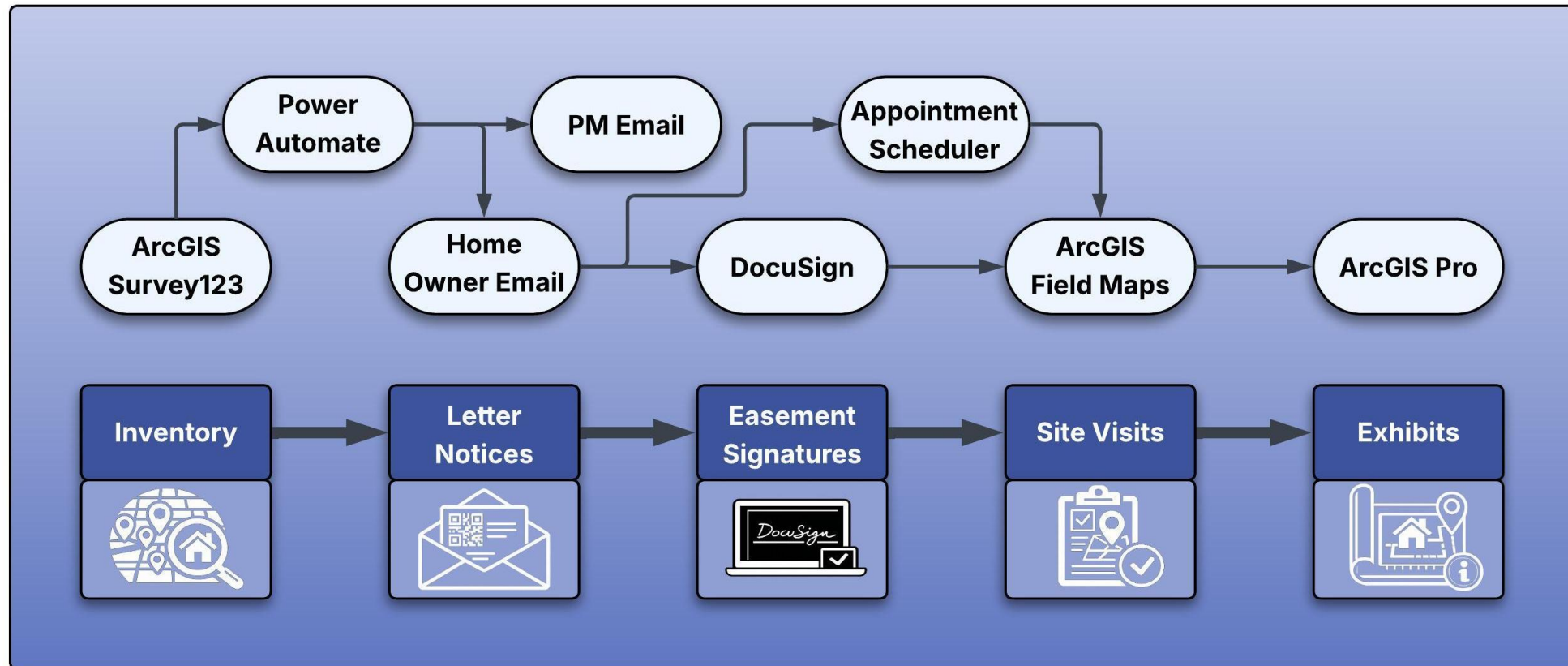
LSL Inventory: Reporting

- All inventories fed into the UMN Lead Service Map via view layers for each city.
- Provided public-facing platform for sharing inventory results

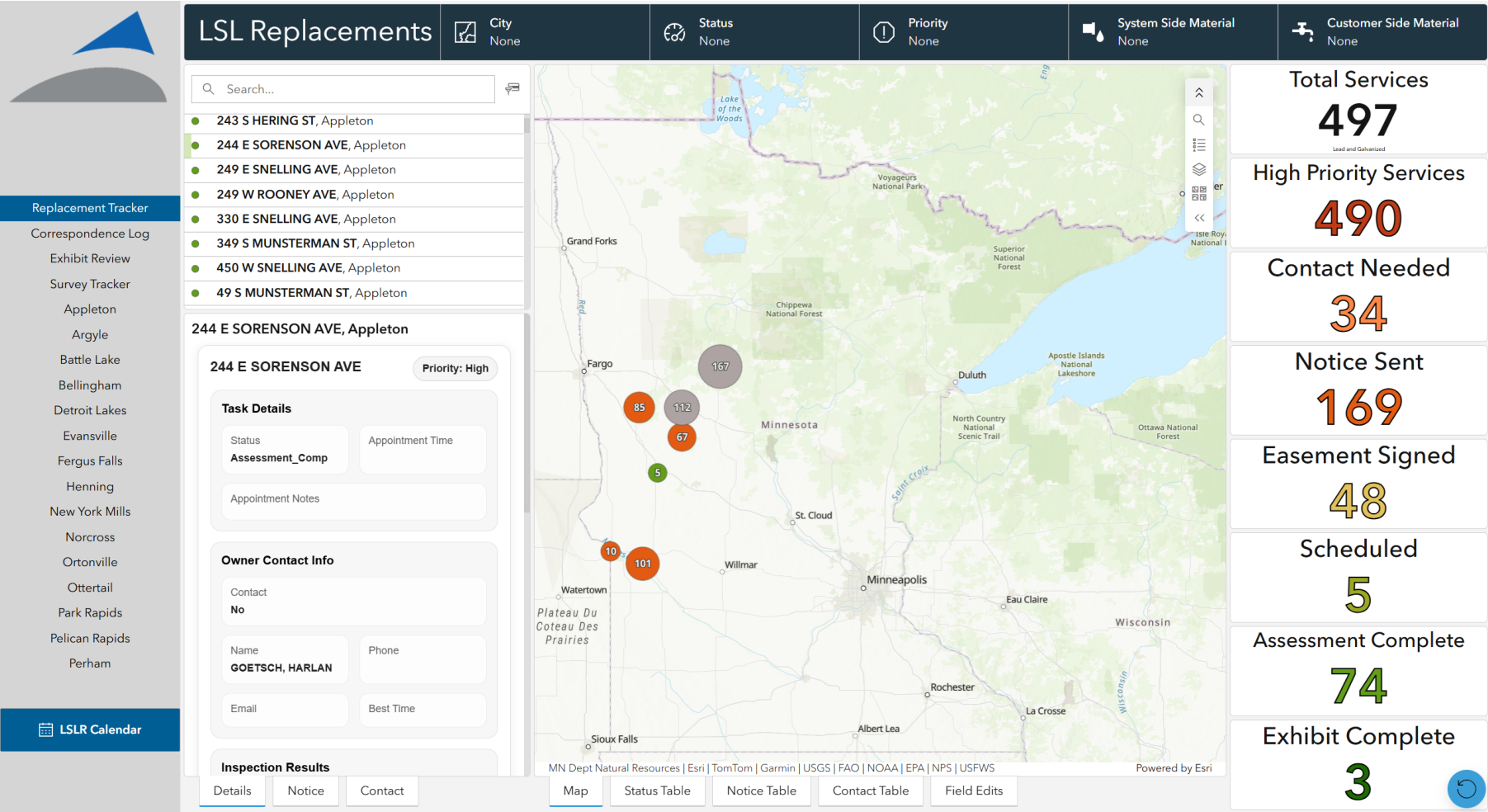


LSL Replacement

- Replacement coordination is the focus of 2026
- GIS was used as the central link between outreach, easement collection, scheduling, field verification, and exhibits
- This workflow connected Survey123, Power Automate, DocuSign, Field Maps, and ArcGIS Pro into one process



LSL Replacement: Dashboards



- Replacement dashboards tracked outreach, signatures, scheduling, assessment status, and exhibits
- PMs could monitor progress by city and quickly identify priorities
- GIS provided a real-time view of where each property stood in the process

LSL Replacement: Homeowner Outreach

- Each community used its own easement agreement
- GIS-based replacement inventories were used to identify eligible homes for outreach
- Survey123, Power Automate, and DocuSign helped streamline contact collection and digital agreement signing



Lead Service Line Replacement Interest Form

This form helps the city understand your interest in participating in the Lead Service Line Replacement program.

Note:

If you decline to participate in the lead service line replacement program at this time, additional LSLR funding may become available in future years, though timing and availability cannot be confirmed.

Interest*

Are you interested in having your home assessed for the lead service line replacement program?

-Please select-

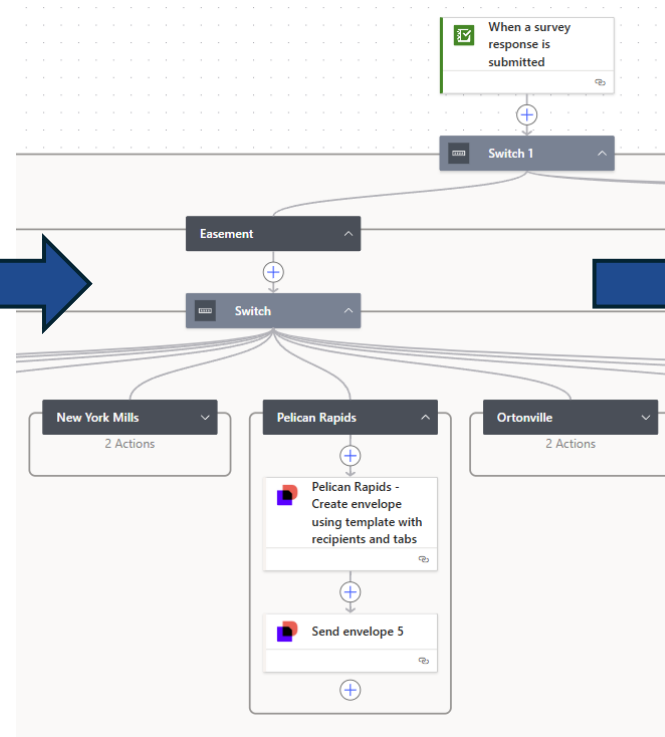
Yes, I am interested and would like to sign an easement agreement.

Yes, I am interested but would like more information.

No, I am not interested.

Submit

Apex
Engineering Group



WATER SERVICE REPLACEMENT AGREEMENT

The purpose of this Agreement is to set forth the terms and conditions pursuant to which the City of Park Rapids (the "City") will provide a new private water service to the undersigned owner's property (the "Property") at:

Property Address	Park Rapids	Minnesota	56470
	City	State	Zip

The new private water service will replace the existing lead or galvanized service. This is being done as part of the Lead Service Line Replacement program (Minn. Stat. § 446A.077) and is an action of the City.

CONSIDERATION:

This Agreement anticipates the work to be performed by the City of Park Rapids staff and/or by a contractor selected by the City of Park Rapids. Each party agrees that the promises made by the other party in this Agreement and the benefits to be derived from their execution are full and sufficient consideration for entering into this Agreement.

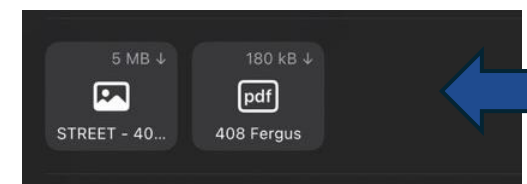
TERM:

The term of this Agreement shall commence on the date of the last signature to this Agreement and shall terminate after completion of the Work and expiration of the City's construction contractor's one (1) year warranty period. The date of the completion of the Work shall be determined by the City.

THE CITY OF PARK RAPIDS RESPONSIBILITIES:

1. Correct/replace the water meter setting and/or meter as necessary at no cost to the undersigned owner of the Property and building (the "Owner"). The Owner is responsible for whatever modifications to the plumbing are required to hook up to the water meter if the water meter is being moved.

2. Install a new water service on the Property constructed by the City of Park Rapids and/or a private contractor(s) at the expense of the City of Park Rapids on the service line of the water meter, between the public right-of-way line and the water meter on the Property (the "new private water service"). The existing private water service and meter may be removed or disconnected and abandoned.



LSL Replacement: Site Visits

- Homeowners were sent an online scheduler to select appointment times
- Scheduled visits were tied back to the GIS workflow for project tracking

Water Service Inspection on April 9th and 10th



Apex Engineering Group
Organizer


45 minutes

Chicago (GMT-05)

▼

Lead Service Line Home Inspection

During this visit, the field staff member will need access inside your home to:

- The water meter
- The electrical panel/breaker box

The inspection is brief and non-destructive (no digging). Please ensure an adult (18+) is home to provide entry, and that the meter area and electrical panel are accessible and not blocked by stored items.

April 2026

<

>

Fri, Apr 10

9:45 am

10:30 am

11:15 am

12:00 pm

12:45 pm

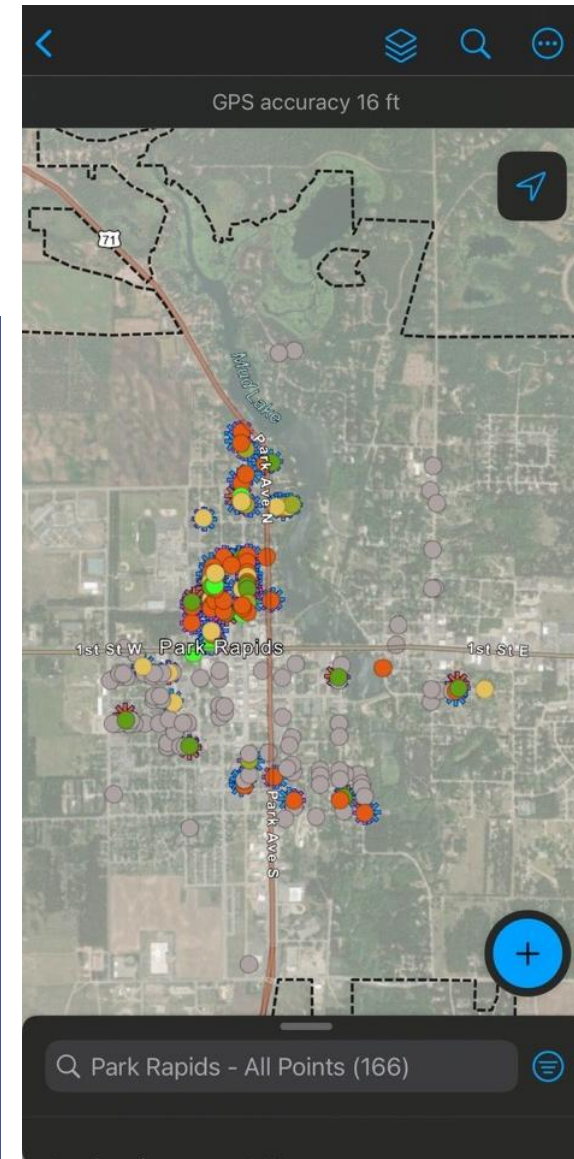
3:00 pm

3:45 pm

4:30 pm

5:15 pm

SU	MO	TU	WE	TH	FR	SA
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		



Park Rapids - All Points (166)

Today · 5 items

**608 PARK AVE N # 5 ,
4/9/2026, 11:15 AM**

608 Park Ave N, Park
Rapids, Minnesota

160.5 mi

**806 MAIN AVE N, 4/9/2026,
1:30 PM**

806 Main Ave N, Park
Rapids, Minnesota

160.7 mi

**214 MAIN AVE N, 4/9/2026,
2:15 PM**

210 Main Ave S, Park
Rapids, Minnesota

160.3 mi

**510 MAIN AVE S, 4/9/2026,
3:00 PM**

510 Main Ave S, Park
Rapids, Minnesota

159.9 mi

**610 MAIN AVE N, 4/9/2026,
4:30 PM**

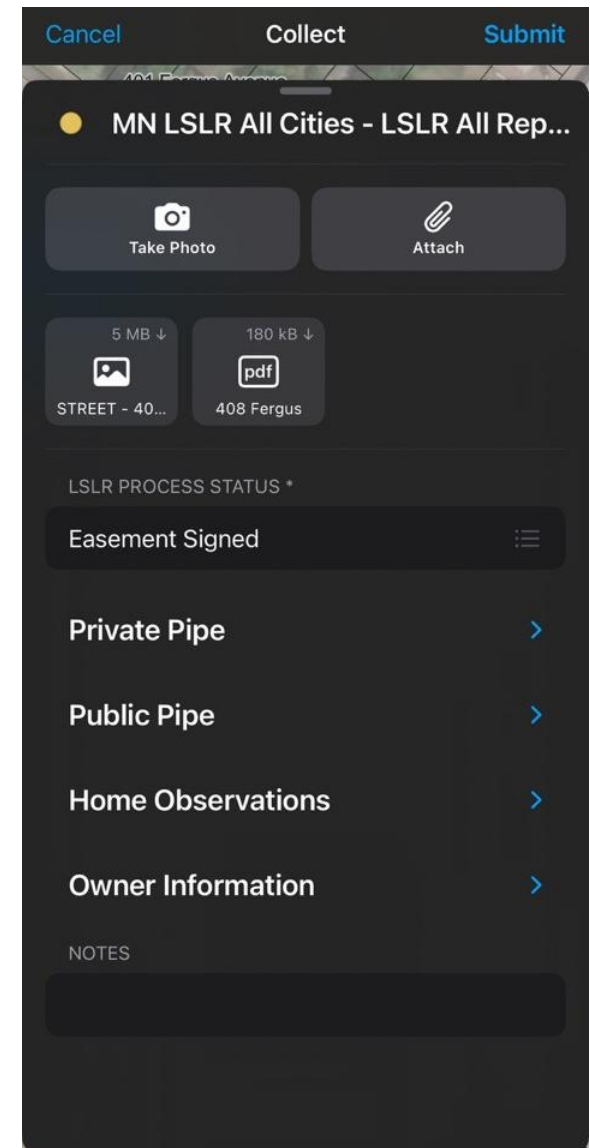
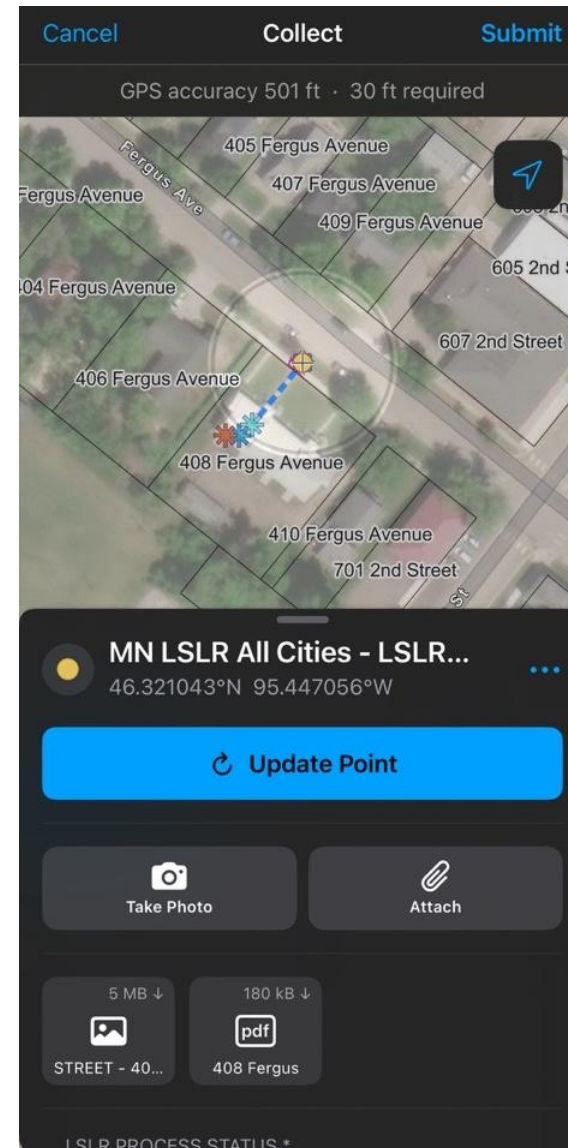
610 Main Ave N, Park
Rapids, Minnesota

160.6 mi

Due date: None · 161 items

LSL Replacement: Site Visits

- Field Maps was used to document site visits, meter locations, homeowner information, and confirmed materials
- Photos and notes were attached directly to each property record



LSL Replacement: Exhibits

- GIS was used to create property-specific exhibits after field review
- Exhibits summarized site conditions, replacement needs, and supporting documentation (site images)
- These materials helped organize the next phase of contractor bidding and project planning



LSL Replacement: Exhibits

The interface shows a map of a residential area with several properties marked. A blue line indicates a proposed service line. The list of exhibits on the right includes:

- 804 COMO ST, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)
- 805 COMO ST, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)
- 201 RIVERSIDE AVE, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)
- 710 FOREST AVE S, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)
- 702 RIVERSIDE AVE, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)
- 711 WASHINGTON AVE S, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)
- 212 PARK AVE N, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)

The detailed exhibit form on the right includes sections for:

- Settings
- LSLR All Replacement
- Exhibit Status (Created)
- General Information
- Owner Information
- Assessment Appointment
- Assessment Information
- Notice_Response (No records were found)
- + Add record

Notes: 3/6/26 not home

- PMs could verify each property exhibit before approval
 - View proposed meter, service line, removals/replacements
 - View street view of home for context
- The result was a more connected workflow from inventory through replacement preparation

LSL Replacement: Exhibits

Search Address

EXHIBIT STATUS

804 COMO ST, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)

805 COMO ST, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)

201 RIVERSIDE AVE, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)

710 FOREST AVE S, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)

702 RIVERSIDE AVE, Park Rapids (Status: Assessment_Comp - Priority: High - Exhibit: Created)

711 WASHINGTON AVE S, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)

212 PARK AVE N, Park Rapids (Status: Notice_Sent - Priority: High - Exhibit: Created)

Total: 62 | Selection: 1

< Edit feature

Settings

LSLR All Replacement

Exhibit Status

Created

General Information

Owner Information

Assessment Appointment

Assessment Information

Notice_Response

No records were found.

+ Add record

Notes: 3/6/26 not home

- PMs could verify each property exhibit before approval
 - View proposed meter, service line, removals/replacements
 - View street view of home for context
- The result was a more connected workflow from inventory through replacement preparation

Summary

- GIS is supporting the entire project process, from inventory through replacement preparation
- Dashboards, Survey123, Power Automate, DocuSign, and Field Maps were connected into one workflow
- Small communities benefited from better tracking, communication, and digital records
- GIS worked best here not just as a mapping tool, but as the framework for project coordination



Efficiencies

Communication

Data Access

Standardization

LSL Inventory and Replacement

Thank You